

BookMyShow Clone Application

End-to-End DevOps Deployment with CI/CD, Kubernetes and Monitoring

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This project demonstrates a complete DevOps lifecycle for deploying a BookMyShow-like web application. It covers containerization, CI/CD automation, Kubernetes orchestration, security scanning, email notifications and real-time monitoring, all implemented on AWS cloud infrastructure.

- The goal was to build a production-style deployment pipeline that is automated, scalable, secure, and observable.

 [EC2](#) > [Instances](#) > [i-06c968d311a07602a](#) 

Instance summary for i-06c968d311a07602a (BMS - Server) [Info](#)

 [Connect](#) [Instance state ▼](#) [Actions ▼](#)

Updated less than a minute ago

Instance ID  i-06c968d311a07602a	Public IPv4 address  13.51.234.133 open address ↗	Private IPv4 addresses  172.31.7.143
IPv6 address -	Instance state  Running	Public DNS  ec2-13-51-234-133.eu-north-1.compute.amazonaws.com open address ↗
Hostname type IP name: ip-172-31-7-143.eu-north-1.compute.internal	Private IP DNS name (IPv4 only)  ip-172-31-7-143.eu-north-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type m7i-flex.large	AWS Compute Optimizer finding  Opt-in to AWS Compute Optimizer for recommendations. Learn more ↗
Auto-assigned IP address  13.51.234.133 [Public IP]	VPC ID  vpc-04fbb5dfae5f66a87 ↗	Auto Scaling Group name
IAM Role	Subnet ID  subnet-0a2c07f604d4...	

- Add all the necessary inbound rules.

EC2 > Security Groups > sg-00f4cb2a061c92101 - launch-wizard-19

ⓘ Inbound security group rules successfully modified on security group (sg-00f4cb2a061c92101 | launch-wizard-19)
 ▶ Details

Owner 787169320097	Inbound rules count 8 Permission entries	Outbound rules count 1 Permission entry
-----------------------	---	--

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (8)

Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-080c8a80babce3f72	IPv4	Custom TCP	TCP	3000 - 10000	0.0.0.0/0
☐	-	sgr-0faa7da98701b2ec1	IPv4	Custom TCP	TCP	30000 - 32767	0.0.0.0/0
☐	-	sgr-0b37d0eaf97d66a5f	IPv4	Custom TCP	TCP	6443	0.0.0.0/0
☐	-	sgr-0743fa6ad3e9167df	IPv4	SMTP	TCP	25	0.0.0.0/0
☐	-	sgr-0d9b106a1e08d1c62	IPv4	SMTPS	TCP	465	0.0.0.0/0
☐	-	sgr-0e56f092426c54484	IPv4	SSH	TCP	22	0.0.0.0/0
☐	-	sgr-080cd80d0e57e4b88	IPv4	HTTP	TCP	80	0.0.0.0/0
☐	-	sgr-0870fcb74f602db96	IPv4	HTTPS	TCP	443	0.0.0.0/0

- Install Jenkins, Docker, Trivy and SonarQube to complete the DevOps toolchain

```
ubuntu@ip-172-31-7-143:~$ docker --version
Docker version 29.2.0, build 0b9d198
ubuntu@ip-172-31-7-143:~$ sudo chmod 666 /var/run/docker.sock
ubuntu@ip-172-31-7-143:~$ docker images

IMAGE ID DISK USAGE CONTENT SIZE EXTRA
ubuntu@ip-172-31-7-143:~$ docker login -u poojaprakash08

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-7-143:~$
```

```
ubuntu@ip-172-31-7-143:~$ trivy --version
Version: 0.68.2
ubuntu@ip-172-31-7-143:~$
```

```
ubuntu@ip-172-31-7-143:~$ docker run -d --name sonar -p 9000:9000 sonarqube:lts-communit
y
Unable to find image 'sonarqube:lts-community' locally
lts-community: Pulling from library/sonarqube
4f4fb700ef54: Pull complete
eb27e3a98da1: Pull complete
5d5a1fad7028: Pull complete
c7ad1fe61e07: Pull complete
f3929ce9ef98: Pull complete
e24a8b9e652f: Pull complete
60d98d907669: Pull complete
1df735f481ad: Pull complete
ce2bd68be4fc: Download complete
600f318704dc: Download complete
Digest: sha256:f709975ab31d2d08f5a3ae2dc73a31ee011afc8cf28845082c17c55d45df9df5
Status: Downloaded newer image for sonarqube:lts-community
e6d3218add6f11c41b4a23e7b783996688e028705a0f9a01726593d0446c558f
ubuntu@ip-172-31-7-143:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	Info →	U	In Use
sonarqube:lts-community	f709975ab31d	1.02GB	398MB	U		

```
ubuntu@ip-172-31-7-143:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
e6d3218add6f	sonarqube:lts-community	"/opt/sonarqube/dock..."	27 seconds ago	Up 22 seconds

```
0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp sonar
ubuntu@ip-172-31-7-143:~$
```

- AWS IAM users and roles are configured securely to manage Kubernetes resources

[IAM](#) > [Users](#) > eks-user

eks-user [Info](#)

[Delete](#)

Summary

ARN
[arn:aws:iam::787169320097:user/eks-user](#)

Created
January 27, 2026, 16:53 (UTC+05:30)

Console access
[Enabled without MFA](#)

Last console sign-in
[Never](#)

Access key 1
[Create access key](#)

[Permissions](#)

[Groups](#)

[Tags](#)

[Security credentials](#)

[Last Accessed](#)

Permissions policies (7)

[Refresh](#) [Remove](#) [Add permissions](#)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type
All types

[<](#) [1](#) [>](#) [Settings](#)

☐	Policy name ↗	Type	Attached via ↗
☐	AmazonEC2FullAccess	AWS managed	Directly
☐	AmazonEKS_CNI_Policy	AWS managed	Directly
☐	AmazonEKSClusterPolicy	AWS managed	Directly
☐	AmazonEKSWorkerNodePolicy	AWS managed	Directly
☐	AWSCloudFormationFullAccess	AWS managed	Directly
☐	eks-inline-policy	Customer inline	Inline
☐	IAMFullAccess	AWS managed	Directly

```
ubuntu@ip-172-31-7-143:~$ vi aws.sh
ubuntu@ip-172-31-7-143:~$ cat aws.sh
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
sudo apt install unzip
unzip awscliv2.zip
sudo ./aws/install
ubuntu@ip-172-31-7-143:~$ sudo chmod +x aws.sh
ubuntu@ip-172-31-7-143:~$ ls
aws.sh  docker.sh  jenkins.sh  trivy.sh
ubuntu@ip-172-31-7-143:~$ ./aws.sh
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 63.7M  100 63.7M    0     0   310M      0  --:--:-- --:--:-- --:--:--  311M
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
ubuntu@ip-172-31-7-143:~$ aws --version
aws-cli/2.33.7 Python/3.13.11 Linux/6.14.0-1018-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-7-143:~$ aws configure
AWS Access Key ID [None]: AKIA30RXSTSQR5X27MUY
AWS Secret Access Key [None]: P6xcmPKjvH/nMbxcUmqoYmC4yd9tCxEbAT9r1xs8
Default region name [None]: eu-north-1
Default output format [None]: json
ubuntu@ip-172-31-7-143:~$
```

- An **EKS (Elastic Kubernetes Service)** cluster is created for container orchestration

```
ubuntu@ip-172-31-7-143:~$ vi kubectl.sh
ubuntu@ip-172-31-7-143:~$ cat kubectl.sh
curl -o kubectl https://amazon-eks.s3.us-west-2.amazonaws.com/1.19.6/2021-01-05/bin/linux/amd64/kubectl
chmod +x ./kubectl
sudo mv ./kubectl /usr/local/bin
kubectl version --short --client
ubuntu@ip-172-31-7-143:~$ sudo chmod +x kubectl.sh
ubuntu@ip-172-31-7-143:~$ ls
aws  aws.sh  awscliv2.zip  docker.sh  jenkins.sh  kubectl.sh  trivy.sh
ubuntu@ip-172-31-7-143:~$ ./kubectl.sh
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 57.4M  100 57.4M    0     0  12.2M      0  0:00:04  0:00:04 --:--:--  12.2M
Client Version: v1.19.6-eks-49a6c0
ubuntu@ip-172-31-7-143:~$
```

```
ubuntu@ip-172-31-7-143:~$ eksctl create cluster --name=poojap-eks \
--region=eu-north-1 \
--zones=eu-north-1a,eu-north-1b \
--version=1.30 \
--without-nodegroup
2026-01-27 12:13:22 [i] eksctl version 0.221.0
2026-01-27 12:13:22 [i] using region eu-north-1
2026-01-27 12:13:22 [i] subnets for eu-north-1a - public:192.168.0.0/19 private:192.168.64.0/19
2026-01-27 12:13:22 [i] subnets for eu-north-1b - public:192.168.32.0/19 private:192.168.96.0/19
```

Create Webhook

All fields marked with * are required

Name *



URL *



Server endpoint that will receive the webhook payload, for example:
"http://my_server/foo". If HTTP Basic authentication is used, HTTPS is recommended to avoid man in the middle attacks. Example:
"https://myLogin:myPassword@my_server/foo"

Secret

If provided, secret will be used as the key to generate the HMAC hex (lowercase) digest value in the 'X-Sonar-Webhook-HMAC-SHA256' header.

[Create](#)[Cancel](#)

- Code quality is analyzed using SonarQube



Jenkins

/ [Manage Jenkins](#)



/ [System](#)



SonarQube installations

[List of SonarQube installations](#)

Name

Server URL

Default is [http://localhost:9000](#)

Server authentication token

SonarQube authentication token. Mandatory when anonymous access is disabled.

- Email notifications are sent for Successful builds and failed builds

 **Jenkins** / Manage Jenkins ▾ / System ▾

E-mail Notification

SMTP server

smtp.gmail.com

Default user e-mail suffix ?

Advanced ^  Edited

☒ Use SMTP Authentication ?

User Name

poojaprakash1808@gmail.com

 For security when using authentication it is recommended to enable either TLS or SSL

Password

.....

☒ Use SSL ?

- Configure the SonarQube and Docker in Jenkins.
- Add credentials for SonarQube, docker and Email ID.

Global credentials (unrestricted) + Add Credentials

Credentials that should be available irrespective of domain specification to requirements matching.

 Sonar-token Sonar-token - Sonar-token	 ...
 poojaprakash08/***** (docker) docker - docker	 ...
 poojaprakash1808@gmail.com/***** (email-creds) email-creds - email-creds	 ...

```
ubuntu@ip-172-31-7-143:~$ eksctl utils associate-iam-oidc-provider \
  --region eu-north-1 \
  --cluster poojap-eks \
  --approve
2026-01-27 19:03:33 [i] will create IAM Open ID Connect provider for cluster "poojap-eks" in "eu-north-1"
2026-01-27 19:03:33 [✓] created IAM Open ID Connect provider for cluster "poojap-eks" in "eu-north-1"
ubuntu@ip-172-31-7-143:~$
```

```
ubuntu@ip-172-31-7-143:~$ eksctl create nodegroup --cluster=poojap-eks \
  --region=eu-north-1 \
  --name=node2 \
  --node-type=t3.medium \
  --nodes=3 \
  --nodes-min=2 \
  --nodes-max=4 \
  --node-volume-size=20 \
  --ssh-access \
  --ssh-public-key=linux1 \
  --managed \
  --asg-access \
  --external-dns-access \
  --full-ecr-access \
  --appmesh-access \
  --alb-ingress-access
```

- A Jenkins pipeline is created to automate the entire workflow

 **Jenkins** / All ▾ / New Item 🔍 ⚙️ 👤

New Item

Enter an item name

BMS-App

Select an item type

**Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

**Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

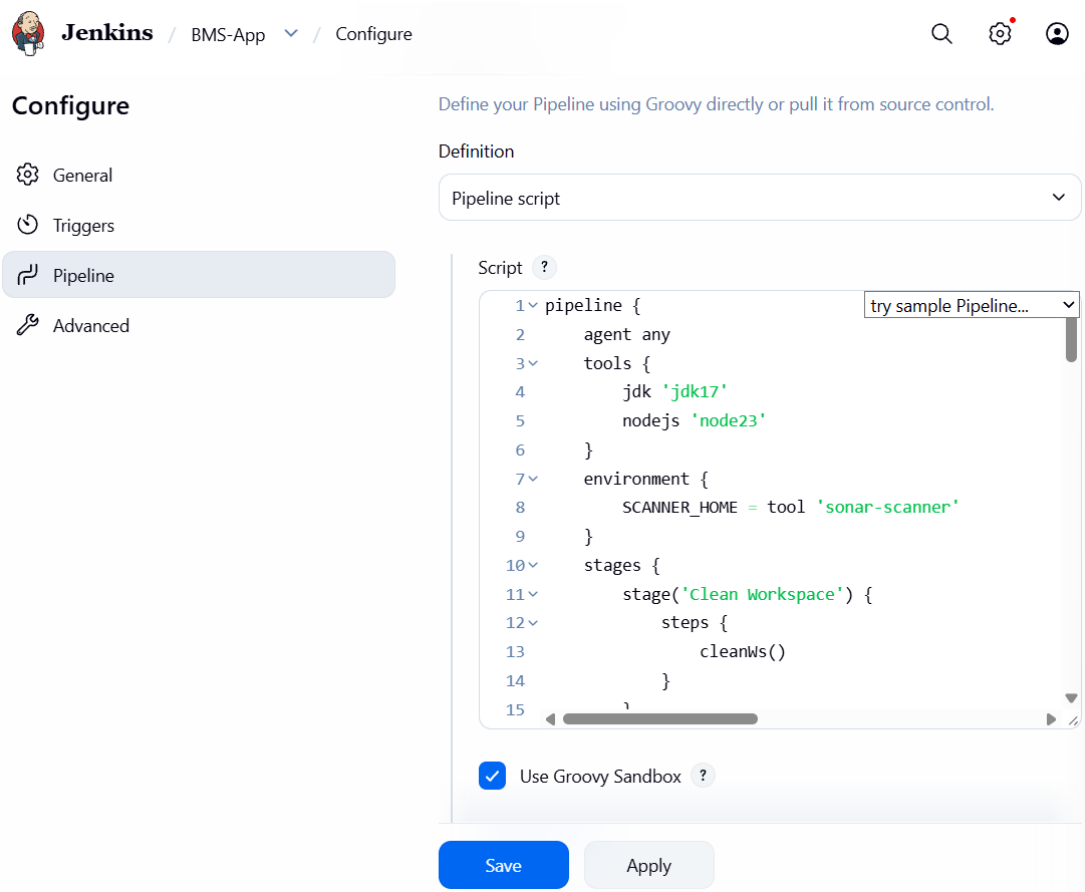
**Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

**Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

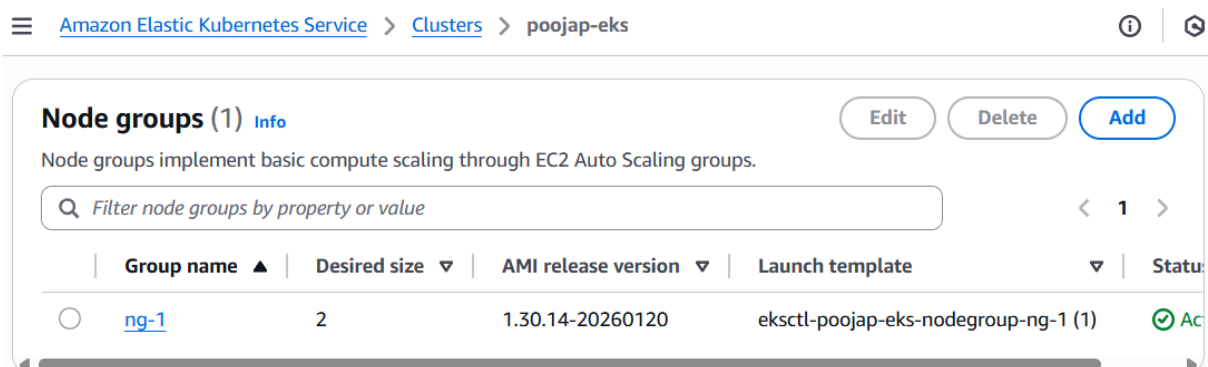
**Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different

OK

- Initially, the application is deployed as a Docker container



- The pipeline is extended to deploy the application into **Kubernetes**:
 - Application runs as pods inside the EKS cluster
 - Kubernetes services expose the application
 - Jenkins interacts directly with the EKS cluster



sonarqube
Projects
Issues
Rules
Quality Profiles
Quality Gates
Administration

?
Search for projects...
A

My Favorites
All

Search by project name or key
Create Project

1 project(s)
Perspective: Overall Status
Sort by: Name

Quality Gate
Passed 1
Failed 0

Reliability (Bugs)
A A rating 0
B B rating 0

BMS Passed
Last analysis: 33 minutes ago

Bugs 18 D
Vulnerabilities 0 A
Hotspots Reviewed 0.0% E
Code Smells 116 A
Coverage 0.0%
Duplications 1.4%
Lines 5.8k S

JavaScript...

Jenkins / BMS-App

Status

BMS-App
Add description

Changes
Build Now
Configure
Delete Pipeline
Full Stage View
SonarQube
Stages
Rename
Pipeline Syntax

	Declarative: Tool Install	Clean Workspace	Checkout from Git	SonarQube Analysis	Quality Gate	Install Dependencies	OWASP FS Scan	Trivy FS Scan	Docker Build & Push	Deploy to Container	Declarative: Post Actions
stage times: ~15min 15s)	173ms	334ms	1s	15s	406ms	1min 47s	8min 11s	5s	4min 43s	6s	397ms
	173ms	334ms	1s	15s	406ms (paused for 2s)	1min 47s	8min 11s	5s	4min 43s	6s	397ms

SonarQube Quality Gate

BMS Passed

server-side processing: Success

Latest Dependency-Check

Not secure 13.51.234.133:3000

bookmyshow
Search for Movies, Events, Plays,
Select City
Sign In

Movies
Stream
Events
Plays
Sports
Activities
Fanhood
Buzz
Listyourshow
Corporates
Offers
Cards

Gift A Cycle
Enable • Empower • Enrich
Helping 200 girls from Alibaug Taluka Cycle to Success
#BookMyShowCares
Click Here To Gift

MASALA SANDWICH
3rd April, 2021

Recommended Movies
See all

The Best of Entertainment

- A separate monitoring server is set up for:
 - **Prometheus** collects metrics from Jenkins and Application nodes.
 - **Node Exporter** provides system-level metrics

[illegible]

← → ↻ Not secure 13.62.225.216:9090/targets?search= ☆ 🌧️ 📄 📄 📄 📄

☰ Prometheus ⚙️ 🌙 🌕

Targets

All scrape pools ▾ All Unhealthy Collapse All 🔍 Filter by endpoint or labels

☒ Unknown
 ☒ Unhealthy
 ☒ Healthy

jenkins (1/1 up) [show more](#)

node_exporter (1/1 up) [show more](#)

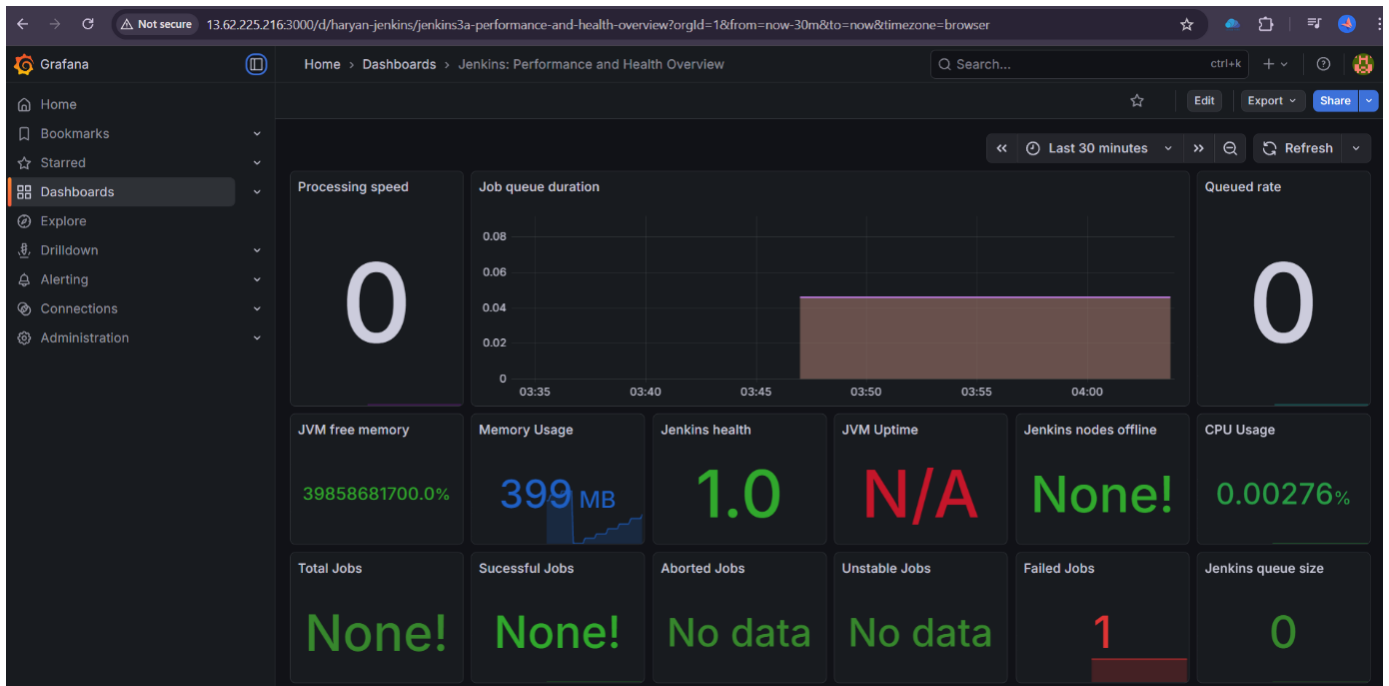
prometheus (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	5.765s ago	3.110ms	

- Grafana visualizes performance data using dashboards

```
ubuntu@ip-172-31-0-191:~$ sudo systemctl start grafana-server
ubuntu@ip-172-31-0-191:~$ sudo systemctl status grafana-server
● grafana-server.service - Grafana instance
   Loaded: loaded (/lib/systemd/system/grafana-server.service; enabled; vendor preset:
   Active: active (running) since Tue 2026-01-27 22:21:23 UTC; 9s ago
     Docs: http://docs.grafana.org
    Main PID: 15407 (grafana)
      Tasks: 12 (limit: 4556)
     Memory: 158.3M
        CPU: 2.466s
    CGroup: /system.slice/grafana-server.service
            └─15407 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana

Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=installer.fs t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=plugins.registration t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=plugin.backgroundinstaller t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=plugin.backgroundinstaller t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=plugin.installer t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=installer.fs t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=plugins.registration t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=plugin.backgroundinstaller t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=plugin.backgroundinstaller t=2026-01-27T22:21:30.000Z
Jan 27 22:21:30 ip-172-31-0-191 grafana[15407]: logger=plugin.backgroundinstaller t=2026-01-27T22:21:30.000Z
Jan 27 22:21:32 ip-172-31-0-191 grafana[15407]: logger=plugin.installer t=2026-01-27T22:21:32.000Z
lines 1-21/21 (END)
```

❖ This project showcases a complete real-world DevOps implementation by automating application delivery from code commit to production deployment with continuous monitoring. It highlights the practical use of CI/CD, containerization, Kubernetes and observability tools to build a scalable, secure and reliable deployment system.